

Operating Systems

2520030042

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SEC-5

WEEK-2

2. Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

Use the `strace` utility to trace all system calls generated by the command `cat sample.txt`. Analyze the sequence of system calls and identify the kernel services involved

Code

```
Ashish@Ashish: ~
GNU nano 8.7.1
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

int main() {
    int source, destination;
    char buffer[1024];
    ssize_t bytesRead, bytesWritten;

    source = open("source.txt", O_RDONLY);
    if (source < 0) {
        perror("Error opening source file");
        return 1;
    }

    destination = open("destination.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);
    if (destination < 0) {
        perror("Error opening destination file");
        close(source);
        return 1;
    }

    while ((bytesRead = read(source, buffer, sizeof(buffer))) > 0) {
        bytesWritten = write(destination, buffer, bytesRead);

        if (bytesWritten != bytesRead) {
            perror("Error writing to destination file");
            close(source);
            close(destination);
            return 1;
        }
    }

    if (bytesRead < 0) {
        perror("Error reading source file");
    }

    close(source);
    close(destination);

    printf("File copied successfully.\n");

    return 0;
}
```

output

```
chmod process completed.
Ashish@Ashish:~$ nano copy.c
Ashish@Ashish:~$ gcc copy.c
Ashish@Ashish:~$ ./a.out
Error opening source file: No such file or directory
Ashish@Ashish:~$ |
```